

Perfect Number

Jojo has a number K and you need to tell whether that number is a Perfect Number or not. A Perfect Number is a number X that is equal to the sum of its positive divisor **excluding** the number itself. For example, 6 is a Perfect Number because the sum of its divisors excluding itself (1, 2, and 3): $1+2+3$ is also 6. Another example is 28, because of the sum of its divisors excluding itself (1, 2, 4, 7, and 14): $1+2+4+7+14$ is also 28.

For this problem, please make a function “**int isPerfect(int K)**” that return 1 if K is a Perfect Number.

Note that for this problem, you can solve it **iteratively**, but it's **encouraged** to use a **recursive solution**. **Do not** include “math.h” library or use any built-in C/C++ function. You need to make your own iterative/recursive function and you do not need any math.h related function in order to solve this problem.

PS: You can declare any other **self-made** iterative/recursive function in order to help you.

Format Input

The input begins with an integer T , indicating the number of test cases. In each test case, there is a number K that Jojo has in one single line.

Format Output

For each test case, output “Case #X:” in the first line followed by “YES” or “NO”.

Constraints

$1 \leq T \leq 100$

$1 \leq K \leq 10000$

Sample Input	Sample Output
4 1233 28 6 9877	Case #1: NO Case #2: YES Case #3: YES Case #4: NO